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2 **Development and Validation of a Need-for-Help and Self-Care Instrument for Postpartum**
3 **Emergency Prevention in Preeclampsia-Eclampsia**

4 **Abstract**

5 **Background and objectives:** Postpartum women with preeclampsia-eclampsia remain at risk of
6 clinical deterioration, yet routine assessment may not systematically integrate clinical warning
7 signs, self-care capacity, and need for support. This study developed and tested a Need-for-Help
8 and Self-Care-based assessment instrument for preventing postpartum emergencies in women
9 with preeclampsia-eclampsia.

10 **Methods:** A research and development design based on the Four-D model was applied. The
11 development phase included expert appraisal by seven validators and a limited usability trial
12 with 20 healthcare providers. Empirical testing involved 30 postpartum women with
13 preeclampsia-eclampsia. Content validity was assessed using Aiken's V. Construct validity was
14 examined using exploratory factor analysis. Reliability was assessed using Cronbach's alpha, and
15 clinical feasibility was evaluated through implementation indicators and provider feedback.

16 **Results:** The overall Aiken's V was 0.89, with domain values of 0.89 for Clinical Warning
17 Signs, 0.92 for Need for Help, and 0.88 for Self-Care. Exploratory factor analysis showed
18 adequate sampling suitability (KMO=0.82; Bartlett's test $p<0.001$) and extracted three factors:
19 Clinical Warning Signs, Self-Care/Activities of Daily Living, and Psychosocial and Support,
20 explaining 62.3% of the total variance. The overall Cronbach's alpha was 0.89. During
21 implementation, 40.0% of patients were identified with warning signs, 75.0% received
22 immediate interventions, 25.0% required intensive care referral, and no maternal deaths
23 occurred. Most providers (85.0%) reported that the instrument was clear and helpful, although
24 item length slowed triage.

25 **Conclusion:** The Need-for-Help and Self-Care-based instrument showed strong validity,
26 reliability, and clinical usability for early detection of postpartum emergencies related to
27 preeclampsia-eclampsia. Further testing in larger and more diverse settings and development of a
28 concise digital version are recommended.

29 **Keywords:** Self Care; Postpartum Period; Pre-Eclampsia; Nursing Assessment; Patient Safety;
30 Reproducibility of Results

31 Introduction

32 Maternal mortality remains an important public health problem in Indonesia. Recent national
33 reports indicate that maternal deaths remain higher than global targets, and preeclampsia-
34 eclampsia continues to be ¹⁵one of the major preventable causes of severe maternal morbidity and
35 mortality (1-3). Although preeclampsia is usually diagnosed during pregnancy, clinical
36 deterioration may continue or newly appear after birth. This creates a need for systematic
37 postpartum monitoring, particularly in facilities where assessment depends on multiple
38 healthcare providers with different levels of experience.

39 Postpartum preeclampsia-eclampsia may progress rapidly and involve several organ systems.
40 Severe hypertension, headache, visual disturbance, seizure, pulmonary edema, renal impairment,
41 disseminated intravascular coagulation, and cardiovascular collapse can occur if warning signs
42 are not recognized early (4-6). Delay in identifying these signs may lead to late referral and
43 delayed treatment. Standardized assessment can help nurses and midwives identify patients who
44 require immediate help, document changes consistently, and communicate patient needs to the
45 clinical team.

46 Previous work developed an assessment ¹⁸instrument for postpartum patients with severe
47 preeclampsia-eclampsia based on Wiedenbach's Need for Help and Orem's Self-Care concepts

(7). However, before such an instrument can be implemented as a routine clinical tool, its content validity, construct validity, reliability, and feasibility must be tested. Instrument validation is essential because unclear wording, weak item relevance, or inconsistent factor structure may reduce the accuracy and usability of clinical assessment tools (8-12).

This study aimed to develop and validate a Need-for-Help and Self-Care-based assessment instrument for postpartum emergency prevention in women with preeclampsia-eclampsia. The specific objectives were to assess content validity, construct validity, reliability, and clinical feasibility during early implementation in postpartum care.

Methods

Study design and setting

This study used a research and development design based on the Four-D model: define, design, develop, and disseminate (8). The study focused on the develop and disseminate stages. The develop stage included expert appraisal, revision, limited trial testing, construct validation, and reliability analysis. The disseminate stage introduced the final instrument to healthcare providers and evaluated preliminary implementation. The study was conducted in the obstetric ward of a public referral hospital in Surabaya, Indonesia, from July to November 2025.

Participants

The development stage involved seven experts: one obstetrician-gynecologist, four midwives, one medical record specialist, and one nursing practitioner/documentation expert. Experts were selected based on clinical or managerial experience in maternal care, postpartum care, documentation, or preeclampsia-eclampsia management. The limited trial involved 20 midwives who provided care for postpartum women. The empirical test involved 30 postpartum

70 women diagnosed with preeclampsia-eclampsia who were admitted to the obstetric ward ⁶ and
71 were able to provide written informed consent.

72 Healthcare providers were included if they provided direct care for postpartum mothers with
73 preeclampsia-eclampsia, had at least one year of experience in maternity or postpartum services,
74 and agreed to participate. Postpartum women were included if they had a documented history of
75 preeclampsia-eclampsia, were cooperative, and were receiving postpartum care. Women with
76 impaired consciousness, severe complications unrelated to preeclampsia-eclampsia, cognitive
77 impairment, or withdrawal of consent were excluded.

78 **Instrument development and data collection**

79 During the define stage, assessment needs were identified through preliminary review of
80 preeclampsia-eclampsia care, existing maternal assessment documents, relevant literature, and
81 interviews with healthcare professionals. During the design stage, indicators and items were
82 developed from Need-for-Help and Self-Care concepts and field findings. The instrument was
83 structured as a checklist covering clinical warning signs, self-care/activities of daily living, and
84 psychosocial and support needs.

85 During the develop stage, content validity was assessed using expert validation forms. Items
86 were revised based on expert feedback about relevance, clarity, and clinical applicability. The
87 limited trial assessed completion time, clarity, ease of use, and general feedback from midwives.
88 The empirical test assessed the instrument's applicability among postpartum women with
89 preeclampsia-eclampsia and generated data for construct validity and reliability testing.

90 **Data analysis**

91 Content validity was calculated using Aiken's V. Items with lower agreement were revised
92 for clarity and clinical relevance. Construct validity was examined using exploratory factor

93 analysis with ² the Kaiser-Meyer-Olkin measure and Bartlett's test of sphericity to determine
94 factorability. Factor loadings of at least 0.40 were considered acceptable. Reliability was
95 examined using Cronbach's alpha, with alpha values of at least 0.70 considered acceptable
96 (16,26,27). ¹⁶ Descriptive statistics were used to summarize participant characteristics, usability
97 indicators, and implementation outcomes.

98 **Ethical considerations**

99 This study was approved by an institutional research ethics committee (No.
100 023/KEPK/07/2025; August 6, 2025). All participants received information about the ¹¹ study
101 objectives and procedures and provided written informed consent before participation.
102 Confidentiality, voluntary participation, and the right to withdraw were ensured throughout the
103 study.

104 **Results**

105 Seven experts participated in the validation stage, with professional experience ranging from
106 3 to 25 years. A limited trial with 20 midwives showed an average completion time of 8.5+/-2.1
107 minutes, a median completion time of 8 minutes, and an ease-of-use score of 4.1+/-0.6 on a 1-5
108 scale. Most providers (85.0%) considered the instrument clear and easy to use.

109 Empirical testing involved 30 postpartum women with preeclampsia-eclampsia. The mean
110 age was 30.2+/-5.1 years. Most participants delivered by cesarean section (60.0%), and 10.0%
111 had a history of chronic hypertension.

112 The content validity analysis showed strong expert agreement. The overall Aiken's V was
113 0.89. Domain values were 0.89 for Clinical Warning Signs, 0.92 for Need for Help, and 0.88 for

Self-Care. Two items with lower agreement were revised to improve wording and clinical alignment.

¹⁷ Exploratory factor analysis indicated that the data were suitable for factor extraction (KMO=0.82; Bartlett's test $p<0.001$). Three factors were extracted and explained 62.3% of the total variance: Clinical Warning Signs (28.5%), Self-Care/Activities of Daily Living (20.1%), and Psychosocial and Support (13.7%). Factor loadings ranged from 0.40 to 0.82. Reliability was strong, with ¹⁹ Cronbach's alpha of 0.89 for the total instrument. Subscale alpha values were 0.88 for Clinical Warning Signs, 0.85 for Self-Care/Activities of Daily Living, and 0.80 for Psychosocial and Support.

During effectiveness testing, 40.0% of patients were identified with warning signs. Among these patients, 75.0% received immediate interventions and 25.0% required intensive care referral. No maternal mortality occurred during the observation period. Providers reported that the instrument helped standardize assessment, although long items and medical terminology could slow triage.

Discussion

The instrument demonstrated strong content validity, indicating that experts considered the items relevant to postpartum preeclampsia-eclampsia assessment. The revised items mainly required clearer wording, which is consistent with instrument development literature emphasizing relevance, comprehensiveness, and comprehensibility as core components of content validity (9-12). Similar maternal and nursing instrument validation studies have also shown that refinement of wording and clinical indicators improves item clarity and practical usability (13,14).

Construct validity was supported by the extraction of three clinically meaningful factors. The first factor, Clinical Warning Signs, explained the largest proportion of variance and included indicators such as severe hypertension and headache. This finding is clinically plausible because hypertensive danger signs remain central to early recognition and management of preeclampsia-eclampsia (15). It also supports the Need-for-Help concept, in which urgent assistance becomes necessary when a patient's condition exceeds her ability to respond independently.

The second factor, Self-Care/Activities of Daily Living, reflects functional ability after childbirth. Functional recovery is important in postpartum assessment because limitations in eating, mobility, hygiene, or self-management may indicate the need for nursing support and closer monitoring. Studies on postpartum functional assessment and self-care-based interventions support the importance of measuring functional ability as part of maternal recovery and complication prevention (17-22).

The third factor, Psychosocial and Support, highlights the role of family support, emotional condition, and coping capacity. Although this factor explained a smaller proportion of variance, psychosocial aspects remain clinically important in postpartum care. Nursing assessment instruments increasingly include patient and family needs, care quality, and support systems as components of comprehensive assessment (23-25).

The instrument also showed strong reliability. The overall Cronbach's alpha of 0.89 indicates good internal consistency and suggests that the items measure related dimensions of postpartum risk and care needs (26,27). This finding is comparable to previous nursing and preeclampsia-related assessment instruments that reported acceptable to high reliability (28,29).

Preliminary implementation suggested that the instrument was clinically useful. It identified warning signs, supported timely intervention, and helped standardize assessment by healthcare

providers. These findings are aligned with evidence that standardized maternal warning systems and postpartum hypertension protocols can improve recognition of deterioration and support clinical decision-making (30-33). The absence of maternal mortality during observation is encouraging, although it should be interpreted cautiously because of the small sample size and short observation period.

This study has limitations. The empirical test involved a relatively small sample from one hospital, which may limit generalizability. The instrument was perceived as useful, but its length and medical terminology may slow triage, particularly in primary care. Future studies should test the instrument in larger and more diverse settings, evaluate predictive accuracy, and develop a shorter or digital version. These steps are consistent with global recommendations to strengthen quality of maternal and newborn care through early detection, standardized assessment, and facility-level implementation (34,35).

Conclusion

The Need-for-Help and Self-Care-based postpartum preeclampsia-eclampsia assessment instrument demonstrated strong content validity, construct validity, reliability, and preliminary clinical usability. The instrument integrates clinical warning signs, self-care/activities of daily living, and psychosocial support into a standardized assessment tool for postpartum care. It may help nurses and midwives recognize deterioration earlier and support timely intervention. Further validation in larger populations and development of a concise digital version are recommended before broader implementation.

Declarations

Funding source(s): No funding was received for this study.

14

Ethics approval and consent to participate: This study was approved by an institutional research ethics committee (No. 023/KEPK/07/2025; August 6, 2025). All participants provided written informed consent before participation.

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Conflict of interest: The authors declare that they have no competing interests.

8

Authors' contributions: Author 1 contributed to conceptualization, methodology, data curation, formal analysis, investigation, and writing the original draft. Author 2 contributed to formal analysis, validation, and writing-review and editing. Author 3 contributed to supervision, methodology, writing-review and editing, and project administration. All authors read and approved the final manuscript.

5

Consent for publication: Not applicable.

Data availability: The data that support the findings of this study are not publicly available due to participant privacy and ethical restrictions. De-identified data may be made available from the corresponding author upon reasonable request and with permission from the relevant ethics committee.

12

Highlights

What is the current knowledge?

- Postpartum women with preeclampsia-eclampsia remain at risk of delayed recognition of clinical deterioration.
- Standardized nursing and midwifery assessment tools can support early detection and timely intervention.

What is new here?

- This study developed a Need-for-Help and Self-Care-based assessment instrument for postpartum preeclampsia-eclampsia.

204 - The instrument showed strong content validity, construct validity, reliability, and preliminary
205 clinical usability.

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Tables

Table 1. Characteristics of expert validators

No.	Profession/expertise	Years of experience
1	Obstetrician-gynecologist	15
2	Midwife	25
3	Midwife	15
4	Midwife	12
5	Midwife	3
6	Medical record specialist	5
7	Nursing practitioner/documentation expert	10

Table 2. Limited trial findings among healthcare providers

Indicator	Result
Number of respondents	20 midwives

Average completion time	8.5+/-2.1 minutes
Median completion time	8 minutes
Ease-of-use score (1-5)	4.1+/-0.6
Respondents rating the instrument clear and easy to use	85.0%
General feedback	Format was easy to understand and practical for field use

307

308 **Table 3. Characteristics of postpartum women in empirical testing (n=30)**

Characteristic	Category/statistic	n	%
Age	Mean+/-SD: 30.2+/-5.1 years	-	-
Mode of delivery	Cesarean section	18	60.0
Mode of delivery	Vaginal delivery	12	40.0
History of chronic hypertension	Present	3	10.0
History of chronic hypertension	Absent	27	90.0

309

310 **Table 4. Content validity of the assessment instrument**

Domain/indicator	Number of items	Aiken's V	Mean	Category	Notes
Clinical Warning Signs	12	0.75-0.95	0.89	High	Item 5 revised
Need for Help (Support)	10	0.80-0.97	0.92	Very good	-
Self-Care (ADL, Functional)	10	0.72-0.94	0.88	Good	Item 12 revised
Overall	32	0.72-0.97	0.89	High	-

311

312 **Table 5. Exploratory factor analysis results**

Factor	Number of items	Example dominant items	Variance explained	Factor loading range
Clinical Warning Signs	11	Blood pressure ≥ 160 mmHg; severe headache	28.5%	0.45-0.82
Self-Care/ADL	10	Eating ability; personal hygiene	20.1%	0.42-0.78
Psychosocial and Support	7	Family support; psychological condition	13.7%	0.40-0.74
Total variance	28 effective items	-	62.3%	-

313

314 **Table 6. Reliability testing results**

Factor/subscale	Number of items	Cronbach's alpha	Category
Clinical Warning Signs	11	0.88	Very good
Self-Care/ADL	10	0.85	Very good
Psychosocial and Support	7	0.80	Good
Overall instrument	32	0.89	Very good

315

316 **Table 7. Preliminary clinical usability and effectiveness indicators**

Indicator	Result	Interpretation
Patients with warning signs	40.0%	Supported early identification
Patients receiving immediate intervention	75.0%	Supported timely clinical response
Patients referred to intensive care	25.0%	Supported escalation of care
Maternal mortality	0 cases	No maternal deaths during observation
Average completion time	8.5+/-2.1 minutes	Practical for hospital use; shorter version needed for primary care
Main obstacles	Long items; difficult medical terminology	Requires simplified wording and glossary

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